Exact AO Density Projections for the Analytic Orbital Hessian Diagonal

Theory and implementation note

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1 Required quantities

Let $U \in \mathbb{R}^{N\times m}$ be the AO-coefficient frame used for the analytic orbital diagonal, where N is the number of AO basis functions. The current complete frame has $m = N$; the identities below also hold for a rectangular frame. For a symmetric, not necessarily positive density D, define

$$J[D]_{\mu\nu} = \sum_{\kappa\lambda} (\mu\nu|\kappa\lambda) D_{\kappa\lambda}, \tag{1}$$

$$K[D]_{\mu\nu} = \sum_{\kappa\lambda} (\mu\kappa|\nu\lambda) D_{\kappa\lambda}. \tag{2}$$

The analytic diagonal requires projected scalar responses

$$j_p[D] = U_p^{\mathsf{T}} J[D] U_p, \qquad k_p[D] = U_p^{\mathsf{T}} K[D] U_p, \tag{3}$$

not the full AO matrices $J[D]$ and $K[D]$. Here U_p is column p of U. For an inactive orbital c, $D_c = U_c U_c^{\mathsf{T}}$. Both $j_p[D_c]$ and $k_p[D_c]$ enter the signed analytic curvature, including the combination $6k_p[D_c] - 2j_p[D_c]$. These contributions cannot be discarded simply because the selected VB structure space is one-dimensional.

For the active orbitals $A \in \mathbb{R}^{N\times n}$, all required density sources have the form $D_t = A W_t A^{\mathsf{T}}$, where W_t is a symmetric active matrix constructed from the accepted spin-free one- and two-particle integral adjoints. The existing untruncated source-span factorization requires $r = \min(n(n+1)/2, 2n)$ basis sources. Its reconstruction matrix recovers the original $2n$ responses and their total one-particle source. No additional numerical rank test or approximation is introduced here.

2 Coulomb responses already available

The accepted integral preparation supplies

$$\mathcal{J}_{p,ab} = (pp|ab). \tag{4}$$

Consequently,

$$j_p[D_t] = \sum_{ab} \mathcal{J}_{p,ab} (W_t)_{ab}. \tag{5}$$

Recomputing $J[D_t]$ through the AO integral graph is unnecessary. Only the active exchange response must still be contracted. The physical response $2j_p[D_t] - k_p[D_t]$ is formed after Eq. (5). The corresponding inactive Coulomb projections are not present in this active tensor and must be evaluated; they can share the exchange traversal.

3 Immediate row projection

At a fixed AO index μ, collect the required exchange rows and inactive Coulomb rows in one matrix,

$$R_\mu(\nu, t) = K[D_t]_{\mu\nu}, \qquad R_\mu(\nu, b+c) = J[D_c]_{\mu\nu}, \tag{6}$$

where b is the current source batch width, and c labels its b_c inactive sources. There are $s = b + b_c \le 2b$ columns. Once this AO row is complete, compute

$$B_\mu = U^\top R_\mu, \qquad d_{pt} = \sum_\mu U_{\mu p}(B_\mu)_{pt}. \tag{7}$$

Equation (7) is exactly Eq. (3), with the contraction order changed. It is not a diagonal approximation to an AO operator. The entire row panel is projected by one matrix multiplication and released before the next owned row. No full AO response, per-channel AO staging image, or second core-Coulomb AO-pair transformation is needed.

Each OpenMP worker owns its AO row panel, projected row panel and partial diagonals. Workers borrow the same densities, integral graph and frame. The numerical row workspace is

$$8T(N + 2m)(b + b_c) \text{ bytes}, \tag{8}$$

where T is the worker count. Including a shared source batch, source-building scratch and required small outputs gives

$$O\big(N^2 b + N^2 + T(N + 2m)b + mP\big), \tag{9}$$

for $P = n_{\text{core}} + r$ sources. In particular, there is no worker-owned $N \times N$ response. Batch width and, when necessary, worker count are chosen within the existing workspace allowance. These choices affect only the exact contraction order and residency, never the physical state space.

The shared density batch remains necessary for this row contraction. Its replacement by a generated or factored source would require a separate algorithm; it is not claimed by Eq. (7). The arithmetic order $O(N^2 mP)$ of the MO projection is also unchanged. The present reductions are redundant Coulomb work, response storage/copying, synchronization and repeated integral visits, not a new asymptotic integral factorization.

4 Integral traversal and batching

For a symmetric AO-pair CSR graph, let e be its directed edge count and d_μ the length of the diagonal pair row (μ, μ). One traversal of all ordered AO output rows visits

$$R = 2e - \sum_\mu d_\mu \tag{10}$$

raw CSR records. If I is the number of permutationally unique integrals, the canonical packed-scatter layout visits $I\lceil P/b_{\text{stream}}\rceil$ records, whereas the projected row layout visits $R\lceil P/b\rceil$. These are record counts, not complete FLOP or wall-time estimates: the layouts exploit integral symmetry differently. Both exact storage orders remain useful, including the canonical stream when the integral graph cannot use 32-bit CSR addresses.

If all P channels fit the row workspace, the final SIMD remainder belongs in the same traversal. Replacing $P = 52$ by a SIMD-aligned batch of 48 would force a second pass over the entire graph without a mathematical need. Alignment of nonfinal batches is useful; alignment must not create another batch when all channels already fit.

For the LOFLEA CAS(12,12) input, $N = 185$, $n_{\text{core}} = 28$, $r = 24$ and $P = 52$. The saved eight-wide canonical run visits 517,740,328 records. A single projected row traversal visits 293,300,715. Actual speed and peak RSS must still be measured under Slurm; the record reduction alone is not a speedup measurement. Numerical validation compares the same signed analytic diagonal against independently transformed four-index integrals, including indefinite sources, repeated AO indices and zero-integral graphs.